

# Atharva Patwardhan

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## Education

**Illinois Institute of Technology**, Chicago, IL May 2026  
MS Applied Mathematics, GPA: 4.00/4.00  
**Indian Institute of Science Education and Research**, Bhopal, India May 2024  
BS - MS Mathematics, GPA: 3.16/4.00

## Work Experience

**Teaching Assistant** August 2025 - May 2026  
Illinois Institute of Technology, Chicago, IL, USA

- Supported instruction for Linear Algebra, Time Series Analysis (ARMA models, stationarity, and forecasting), and Probability and Statistics, reinforcing mathematical foundations critical to quantitative modelling and financial data analysis.
- Led office hours and assisted with grading and assessments, providing rigorous feedback on stochastic processes, matrix methods, and statistical reasoning used in quantitative research and risk modelling.

**Intern- Business Intelligence Unit (BIU)** May 2025- July 2025  
ICICI Prudential Life Insurance, Mumbai, India

- Analyzed key performance indicators (KPIs) driving life insurance purchase decisions, improving process efficiency by 20%.
- Developed a project to detect fraudulent auto insurance claims, enhancing pipeline accuracy by 15%.

**Research Assistant** March 2025 - May 2025  
Illinois Institute of Technology, Chicago, IL, USA

- Served as a Research Assistant under the Summer Undergraduate Research Experience (SURE) program with Prof. Yuhang Ding, contributing to the evaluation and selection of candidates from a pool of over 200 applicants.
- Screened and assessed student applications based on academic performance, skills, and project fit, streamlining the selection process and ensuring a high-quality cohort.

**Graduate Research Assistant** July 2023 - May 2024  
IISER Bhopal, India

- Performed project on **The Theory of Maass Forms** under Dr. Karam Deo Shankhadhar and Prof. Manickam Murugesan, focusing on advanced mathematical concepts.
- Conducted in-depth research on connections between Maass forms and Dirichlet series, applying Rankin-Selberg methods to explore automorphic forms and modular forms theory.

**Summer Research Intern** May 2022 - July 2022  
Fractal Dataminds (FDm), Singapore

- Developed data processing pipelines, handling 1 million+ data points daily, increasing efficiency by 30%.
- Collaborated with a cross-functional team of 8 during coding sprints, contributing code enhancements improving application load times by decreasing server response rates from 2 seconds down to virtually instant loading.
- Solved product requirements for data-driven applications, enhancing success by 20%.

## Project Experience

**Dubins path** Jan 2023 - May 2023  
IISER Bhopal, India

- Performed project on **Bounding the shortest Dubins path through a sequence of points** in a team of 3 focused on the path planning problem for robots and Unmanned Aerial Vehicles (UAVs).
- Improved the lower bound for the problem of finding a shortest path of bounded curvature passing through a sequence of points.

## Skills

**Technical Skills:** Mathematical modeling, Data analysis, Machine learning

**Programming Languages:** Python, SQL, C++, R, Matlab

**Software Tools:** LaTeX, Microsoft Office, Git, Jupyter Notebooks

## Certifications

- Data Science and Machine Learning
- Python
- Statistics for Data Science and Business Analytics

## Relevant Courses

**At IIT:** Time Series, Statistics, Probability, Monte Carlo Methods in Finance, Regression, Mathematical Modelling, Computational Mathematics, Discrete Applied Mathematics.

**At IISER Bhopal:** Complex Analysis, Analytic Number Theory, Algebraic Number Theory, Functional Analysis, Measure & Integration, Analytic Number Theory, Algebraic Number Theory.

## Honors

- Passed qualifying examinations in Statistics and Computational Mathematics at the PhD-level.
- Awarded **DST-INSPIRE** Scholarship by the Government of India, given to the top 1% of science students.
- **1st place** in the Thai Premier Cup International Mathematics Competition, outperforming 300+ participants.

## Volunteer Experience

- Organized and volunteered at IISER Bhopal's Open House 2022, guiding 100+ visitors through science demonstrations.
- Taught and mentored underprivileged children in rural Maharashtra, improving their math skills over a 3-month program.